

06 — Runbook (Run It, Break It, Fix It)

Level 1 — Docker Compose (5 minutes, easiest)

```
# one-time: create .env in project root (see .env values in team vault)
docker-compose up --build -d
docker ps                                # expect ~19 containers
curl http://localhost:8000/api/products  # through Kong - should answer
```

Full user journey by hand:

```
# 1. register
curl -X POST http://localhost:8000/api/users/register \
  -H "Content-Type: application/json" \
  -d '{"name":"Ali","email":"a@a.com","password":"password123"}'
# 2. create product
curl -X POST http://localhost:8000/api/products \
  -H "Content-Type: application/json" \
  -d '{"name":"Phone","price":499.99,"category":"electronics"}'
# 3. add stock
curl -X POST http://localhost:8000/api/inventory/1/restock \
  -H "Content-Type: application/json" -d '{"quantity":10}'
# 4. order it
curl -X POST http://localhost:8000/api/orders \
  -H "Content-Type: application/json" \
  -d '{"userId":1,"shippingAddress":"Pune","items":[{"productId":1,"productName":"Phone","quantity":1,"price":499.99}]}'
# 5. a few seconds later - status should be PAID (the saga ran!)
curl http://localhost:8000/api/orders/1
```

Level 2 — Floci + Kubernetes (the full cloud simulation)

```
docker-compose up -d floci          # start fake AWS
bash scripts/01-vpc.sh               # network
bash scripts/02-alb.sh               # front gate
bash scripts/03-ecr.sh               # build+push images
bash scripts/04-eks.sh               # kubernetes cluster
bash scripts/05-deploy.sh            # deploy everything
kubectl get pods -n ecommerce        # all should be Running
```

Level 3 — CI/CD

```
git push origin develop    # → tests + auto deploy-dev + smoke test
# PR develop→main → merge → approve the production gate in GitHub UI
```

Debug cheat-sheet (the commands that solve 90% of problems)

Symptom	Command	What to look for
Pod not starting	<code>kubectl describe pod</code> <code>-n ecommerce</code>	Events at the bottom: ImagePullBackOff? CreateContainerConfigError (missing Secret)?
App crashing	<code>kubectl logs</code> <code>-n ecommerce</code>	Java stack trace — usually DB not reachable or missing env var
Pod restarts repeatedly	<code>kubectl get pods -n ecommerce</code> (RESTARTS column)	livenessProbe failing → app too slow to start or /health broken
Order stuck PENDING	<code>kubectl logs -l app=inventory-service -n ecommerce</code>	Did inventory consume order.created? Stock available?
404 through Kong	<code>curl localhost:8001/routes</code>	Is the route there? <code>strip_path: false</code> present?
Kafka silence	exec into kafka pod → <code>kafka-topics --list --bootstrap-server localhost:9092</code>	Topics exist?
Everything weird	<code>docker-compose down -v && docker-compose up --build -d</code>	Nuclear option (wipes data)

Golden rules

1. **Logs first.** 90% of answers are in `kubectl logs` / `docker logs`.
2. **Events second.** `kubectl describe pod` events explain why K8s won't start something.
3. Container can't see `localhost` — services reach each other **by name**.
4. Killed a pod and it came back? That's not a bug — that's the reconciliation loop (doc 02).